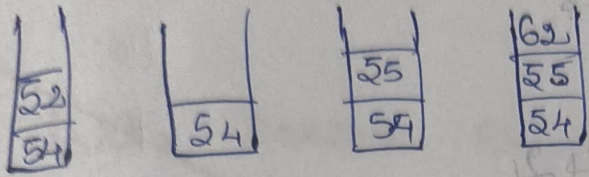


Activity - 2

190572149
M. Debadharshini

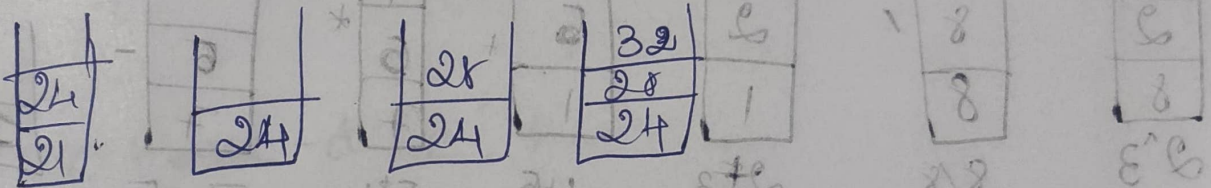
1. Stack



$$S = \text{pop}()$$

$$S = 62$$

2. Queue:



$$Q = \text{dequeue}()$$

$$Q = 24$$

$$S + Q = 62 + 24$$

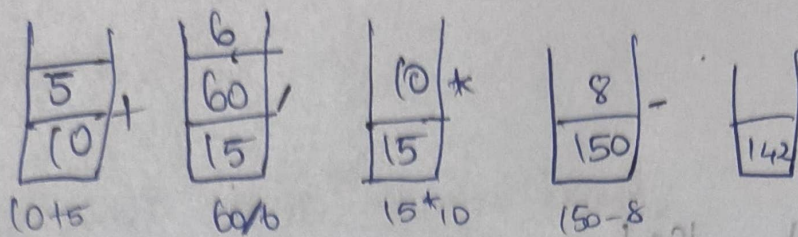
$$= 86$$

$$[a, b, c, d] = [a, b, c, d]$$

To move a → rotate 3 times
To move b → rotate 2 times
To move c → rotate 1 time
To move d → rotate 0 times

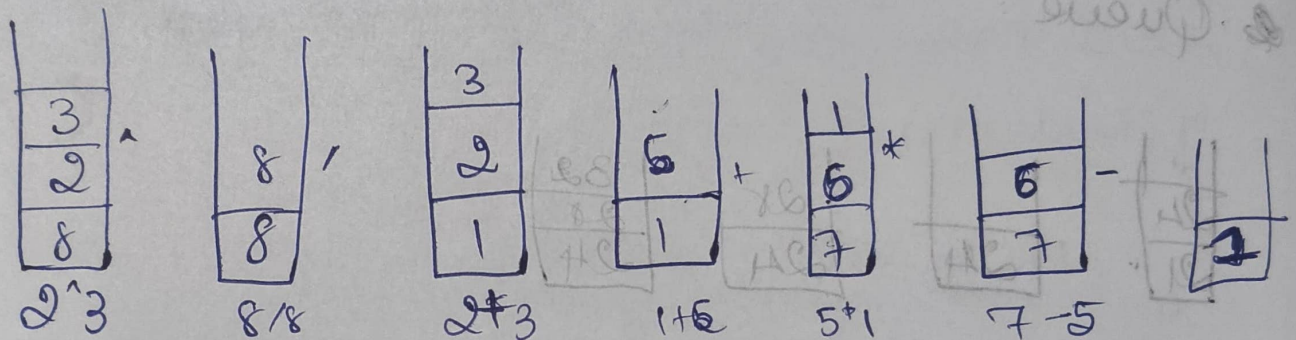
$$d = 1 + 2 + 3 + 4 = 10$$

2. Postfix : $10\ 5\ +\ 60\ 6\ /\ * 8\ -$



$\therefore$ Answer = 142

2. Postfix : $8\ 2\ 3\ ^\wedge /\ 2\ 3\ * + 5\ 1\ * -$



$\therefore$ Answer = 2

4.

Let $Q_1 = [a, b, c, d]$

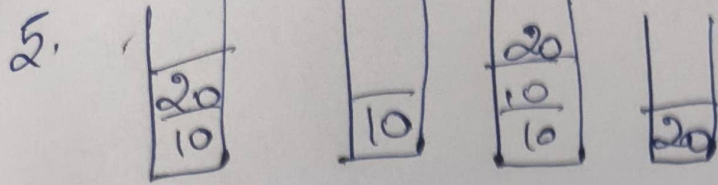
We need $Q_2 = [d, c, b, a]$

To move $d \rightarrow$ rotate Q_1 3 times

To move $c \rightarrow$ rotate Q_1 2 times

To move $b \rightarrow$ rotate Q_1 1 time

$$Q_1 \Rightarrow 3+2+1 = 6$$



Popped values: 20, 20, 10, 10, 20.

6. ~~With reverse operation, stack acts as a queue.~~

~~Eg: Push() - Enqueue
Pop() - Dequeue~~

After reversing,

6. $a + b \times c - d^e^f$.

Postfix: $abc * + def ^ ^ -$